Discriminative Reconstruction for Hyperspectral Anomaly Detection With Spectral Learning

Jie Lei, *Member, IEEE*, Shuo Fang, Weiying Xie, *Member, IEEE*, Yunsong Li, and Chein-I Chang, *Life Fellow, IEEE*

***Abstract*— Recently, autoencoder (AE)-based anomaly detection has drawn considerable interest in hyperspectral image (HSI) analysis. In this article, we propose a novel discriminative reconstruction method for hyperspectral anomaly detection images with spectral learning (SLDR). The proposed algorithm has the following innovations. First, we use the spectral error map (SEM) to detect anomalies because the SEM can preferably reflect the spectral similarity of each pixel between the input and the reconstruction. Second, the loss function of the proposed SLDR model additionally introduces the spectral angle distance (SAD), which constrains the model to generate a reconstruction having greater spectral similarity to the input. Third, a constraint is imposed on the encoder, forcing it to generate latent variables that obey a unit Gaussian distribution, which helps the decoder to reconstruct a better background with respect to the input. Compared with the Reed–Xiaoli (RX), collaborative representation detection (CRD), attribute and edge-preserving filtering-based anomaly detection (AED) and adversarial autoencoder-based anomaly detection (AAE), through two real HSI data sets, the detection performance of the proposed SLDR method is found to be competitive.**

***Index Terms*— Anomaly detection, hyperspectral image (HSI), reconstruction, spectral learning.**

Manuscript received November 3, 2019; revised December 31, 2019 and February 11, 2020; accepted March 18, 2020. Date of publication April 6, 2020; date of current version September 25, 2020. This work was supported in part by the National Natural Science Foundation of China under Grant 61801359, Grant 61571345, Grant 91538101, Grant 61501346, Grant 61502367, and Grant 61701360, in part by the Young Talent Fund of the University Association for Science and Technology in Shaanxi of China under Grant 20190103, in part by the Special Financial Grant from China Postdoctoral Science Foundation under Grant 2019T120878, in part by the 111 Project under Grant B08038, in part by the Fundamental Research Funds for the Central Universities under Grant JB180104, in part by Natural Science Basic Research Plan in Shaanxi Province of China under Grant 2019JQ153, Grant 2016JQ6023, and Grant 2016JQ6018, in part by General Financial Grant from the China Post-Doctoral Science Foundation under Grant 2017M620440, in part by the Yangtse Rive Scholar Bonus Scheme under Grant CJT160102, and in part by the Ten Thousand Talent Program. *(Corresponding author: Weiying Xie.)*

Jie Lei, Shuo Fang, Weiying Xie, and Yunsong Li are with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an 710071, China (e-mail: jielei@mail.xidian.edu.cn; shuofang@stu.xidian.edu.cn; wyxie@xidian.edu.cn; ysli@mail.xidian.edu.cn).

Chein-I Chang is with the Center of Hyperspectral Imaging in Remote Sensing, Information and Technology College, Dalian Maritime University, Dalian 116026, China, also with the Department of Computer Science and Information Engineering, National Yunlin University of Science and Technology, Douliu 64002, Taiwan, also with the Remote Sensing Signal and Image Processing Laboratory, Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore, MD 21250 USA, and also with the Department of Computer Science and Information Management, Providence University, Taichung 02912, Taiwan (e-mail: cchang@umbc.edu).

Color versions of one or more of the figures in this article are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2020.2982406

I. Introduction

HYPERSPECTRAL images (HSIs) provide a wealth of spectral information [1] that is provided by the visible, near-infrared, and shortwave infrared bands in remotely sensed scenes [2]. These spectra are divided into hundreds or even thousands of narrow contiguous spectral bands to capture more accurate measurements of the components of each pixel [3], [4]. Hyperspectral imaging has been widely applied to fields such as target detection [5], [6], anomaly detection [7], change detection [8], image classification [9], [10], band selection [11], [12], and hyperspectral unmixing [13], [14] due to its high spectral resolution. Among these applications, hyperspectral anomaly detection is of particular interest due to its capability of uncovering many unknown material substances [15], [16].

Anomaly detection is a special category of target detection. It can be considered to be a binary classification problem that is designed to classify each pixel in an HSI as either an anomaly or background [17]. Different from the supervised target detection problems that were considered in [18] and [19], anomaly detection is an unsupervised target detection technique that detects anomalies without any prior information of the scene [20]. For HSIs, the goal of anomaly detection is to identify unusual objects that significantly deviate from the typical background with respect to either space or spectrum. For example, the airplanes in an airport background and tanks in a forest background can be judged as anomalies. There are several challenges in hyperspectral anomaly detection. First, there is no prior knowledge of anomalies or backgrounds that is available for data analysis. Second, complex backgrounds hinder detection [21]. Third, the HSI data are usually corrupted by unknown interference due to significantly improved spectral resolutions [22].

In the past years, various hyperspectral anomaly detection algorithms have been developed. The Reed-Xiaoli (RX) detection algorithm [23] is the most classical anomaly detection algorithm. It calculates the Mahalanobis distance between the measured pixels and the background mean by assuming that the probability density function of the background can be modeled as a multivariate Gaussian distribution [24]. By comparing the Mahalanobis distance to a preset threshold, the pixel is determined as a background or an anomaly. The RX detection algorithm has high computational efficiency and excellent detection performance when the background

0196-2892 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:46:56 UTC from IEEE Xplore. Restrictions apply.

distribution of the data set satisfies Gaussian distribution [25]. Depending upon how the background is described, the RX detection algorithm can be implemented in two versions [26]. One is the local RX (LRX) [27], which regards the region surrounding the measured pixel as the background and then uses a dual-window method to detect anomaly targets. The other is the global RX (GRX), which considers the entire image scene as the background. Since all the pixels in an entire image scene or a local region are used to compute the background statistics, the mean and the covariance matrix of the background are easily contaminated by anomalies [28], [29], which may lead to the performance of the RX algorithm being unstable. Because of that, many variants of the RX anomaly detection algorithm have been proposed. For example, a kernel RX (KRX) detection algorithm [30] expands a linear non-Gaussian model in the low-dimensional original data space to a nonlinear Gaussian domain in higher dimensional feature space [31]. The subspace RX (SSRX) detection algorithm [32] employs principal component analysis (PCA) and then applies the RX detection algorithm on limited numbers of PCA bands.

In addition to the statistical approaches described above, representation theory is also applied to anomaly detection algorithms. The representation-based method assumes that HSI pixels can be represented by a dictionary [33]. For instance, collaborative representation detection (CRD) [34] assumes that background pixels can be approximately represented by a linear combination of its neighboring spatial pixels while anomaly pixels cannot, which means that the background pixels and anomaly pixels can be distinguished. The background joint sparse representation detector (BJSRD) [35] measures the length of the matched projection of the tested pixel on the orthogonal complementary background subspace by using the joint sparse representation (JSR) [36] to detect anomalous pixels. The low-rank and sparse representation (LRASR) detector [37] models the background by using a low-rank representation with a sparse regularization term in its coefficients. However, regardless of whether it is a sparse representation-based detector or a CRD algorithm, the performance of the detector is highly correlated with the dictionary that is used. In other words, if the background is contaminated by anomalies, the detection performance may be degraded. Recently, Xie *et al.* [38] proposed an adaptive weighting detector based on the structured tensor and guided filtering (STGF) algorithm for hyperspectral anomaly detection, which achieved better detection performances. In [39], a novel tensor nuclear norm is used to constrain the background tensor to characterize the low-dimensional structure of the background. Wu *et al.* [40] proposed a novel anomaly detection method via global and local joint modeling of background. Moreover, many machine learning or deep learning works focusing on feature learning methods are proposed in [41] and [42].

Recently, autoencoders (AEs) have been used on several occasions for anomaly detection. AEs tend to learn data distribution. Xia *et al.* [43] is the earliest method for detecting anomalies with the reconstruction errors of an AE. Cao *et al.* [44] concatenated several AEs to form a deep-stacked AE that uses the latent representation of an AE as the input of the successive AE. In addition, deep generative models have made significant contributions to unsupervised anomaly detection [45], such as variational autoencoders (VAEs) [46] and generative adversarial networks (GANs) [47]. An and Cho [48] used a VAE for anomaly detection due to its high reconstruction probability and claimed that using a VAE for anomaly detection was easier than using an AE. The GAN-based methods for anomaly detection that were proposed by Zenati *et al.* [49] enabled the network to learn encoders, generators, and discriminators simultaneously in the training process and then use the combination of reconstruction and discrimination losses to determine whether the sample under testing is an anomaly. Kliger and Fleishman [50] applied a similar idea to detect anomalies using a GAN by considering the "fake-class probability" as a score.

Different from previous works, we proposed a novel anomaly detection algorithm based on discriminative reconstruction with spectral learning (SLDR). This model effectively reconstructs the background by learning the spectral information of the input HSI. The spectral error map (SEM) between the input and reconstruction is calculated, where the spectral angle distance (SAD) among the abnormal pixels is large and the SAD among the background pixels is small. That is, the background is suppressed, and any abnormality is highlighted in the SEM. In this way, using the SEM for detection improves the anomaly detection performance. The proposed method works as follows. First, a latent vector is extracted from the input vector of the HSI using the encoder network. A constraint is then imposed on the encoder to force it to generate latent variables that obey a Gaussian distribution, which can help the decoder to better reconstruct the background. Next, the latent vector is decoded to reconstruct the input by a decoder network. Finally, the SEM between the input and the reconstruction is sent to the detector for anomaly detection. The detection performance of the proposed SLDR method is compared with the RX, CRD, attribute and edge-preserving filtering-based anomaly detection (AED), and adversarial autoencoder-based anomaly detection (AAE) through two real HSI data sets experiments. The experimental results demonstrate the effectiveness of the proposed method.

The main contributions of our proposed method can be summarized in three ways.

1) Instead of the reconstruction error, the SEM is employed for anomaly detection, which can better reflect the spectral similarity of each pixel between the input and the reconstruction.
2) The SAD is used as a loss function, which enforces the maximum angular similarity between the input and the reconstruction.
3) The encoder is constrained to produce latent variables that obey a unit Gaussian distribution, which allows the decoding layer to better reconstruct the background that is associated with the input.

The rest of this article is organized as follows. In Section II, the related works are reviewed. Section III describes the proposed SLDR anomaly detection method in detail. The proposed method and other traditional anomaly detectors are

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:46:56 UTC from IEEE Xplore. Restrictions apply.

presented and compared via experiments in Section IV. Finally, the conclusions are drawn in Section V.

II. Related Works

A. AE-Based Anomaly Detection

An AE consists of an encoder and a decoder and can be viewed as an unsupervised learning algorithm. It aims to learn reconstructions that are close to the original input. Given the visible input $\mathbf{x} \in \mathbb{R}^D$, the encoder first maps the input into a latent layer $\mathbf{z} \in \mathbb{R}^B$ as in (1). D and B represent the dimensions of the input layer and latent layer, respectively

$$\mathbf{z} = \sigma(\mathcal{W}_e \mathbf{x} + b_e) \tag{1}$$

where $\mathcal{W}_e$ is the weight matrix, b_e is the bias, and $\sigma(\cdot)$ is the activation of the input units. The latent units are then cast into the output layer $\hat{\mathbf{x}} \in \mathbb{R}^D$ by the decoder, as in the following equation:

$$\hat{\mathbf{x}} = \delta(\mathcal{W}_d \mathbf{z} + b_d) \tag{2}$$

where $\mathcal{W}_d$ is the weight matrix, b_d is the bias, and $\delta(\cdot)$ is the activation of the latent units. The loss function can be defined as the reconstruction error between the input and output units of the AE

$$\mathcal{L}_{\mathrm{AE}} = \|\mathbf{x} - \hat{\mathbf{x}}\|^2. \tag{3}$$

The AE learns to minimize the reconstruction error in the iterative process.

AE-based anomaly detection is a semisupervised learning algorithm that employs the reconstruction error of each tested data as the anomaly score. The data points with high anomaly scores are considered to be anomalies because only the normal instances are used to train the AE. After training, the AE will have better performance when reconstructing normal data while failing to do so with anomaly data that the AE has not encountered.

B. VAE-Based Anomaly Detection

A VAE is a deep-generation model that has been developed in recent years. It is a powerful method for generating samples that are similar to but not the same as the samples in the database. An AE architecture is employed to accomplish a variational approximation of the latent space. Different from a typical AE, it constrains the latent vector $\mathbf{z}$ to be a random variable that is distributed according to a prior distribution $p_\theta(\mathbf{z})$, which is usually a multivariate unit Gaussian $\mathcal{N}(\mathbf{0}, \mathbf{I})$. The true posterior $p_\theta(\mathbf{z}|\mathbf{x})$ is intractable for a continuous latent space $\mathbf{z}$; thus, the variational inference method is introduced to fit another neural network as the approximate posterior $q_\phi(\mathbf{z}|\mathbf{x})$.

The VAE jointly trains the approximated posterior and the generative model by maximizing the evidence lower bound (ELBO). For the input data $\mathbf{x}$, the ELBO is defined as the following equation:

$$\begin{aligned}
&\mathcal{L}_{\mathrm{ELBO}}(\theta, \phi; \mathbf{x}) \\
&\quad = \log p_\theta(\mathbf{x}) - D_{\mathrm{KL}}\left[q_\phi(\mathbf{z}|\mathbf{x}) \| p_\theta(\mathbf{z}|\mathbf{x})\right] \\
&\quad = E_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x}) + \log p_\theta(\mathbf{z}|\mathbf{x}) - \log q_\phi(\mathbf{z}|\mathbf{x})\right] \\
&\quad = E_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x}|\mathbf{z})\right] - D_{\mathrm{KL}}\left[q_\theta(\mathbf{z}|\mathbf{x}) \| p_\theta(\mathbf{z})\right]
\end{aligned} \tag{4}$$

where θ and ϕ are the encoder and decoder parameters, respectively. The second term represents the KL divergence, which measures the difference between the approximate posterior and the prior distribution. Therefore, the objective function of the VAE that is described in (4) weighs the accuracy of the reconstructed input and the fitting degree of the unit Gaussian distribution. The expectation in the above equation can be approximated by a Monte Carlo simulation [51].

AEs define the reconstruction error as the anomaly score in the test phase, while VAEs use the reconstruction probability $E_{q_\phi(z|x)}\left[\log p_\theta(\mathbf{x}|\mathbf{z})\right]$ to detect anomalies. For a new test sample $\mathbf{x}$, the mean and standard deviation vectors are first evaluated with the probabilistic encoder. To estimate the probabilistic anomaly score, VAEs sample $\mathbf{z}$ according to the prior $p_\theta(\mathbf{z})$ L times. Then, the average reconstruction error is calculated as the reconstruction probability. Consequently, VAEs are more robust than traditional AEs in the anomaly detection domain.

VAE has been successfully applied to anomaly detection of traditional images [48], [52], [53]. Because of its powerful representational capability, we apply and develop it to dig intrinsic spectral features from high-dimensional spectral vectors. In particular, the design of network architecture takes full account of the characteristics of HSIs.

III. SLDR Model

By observing the spectral curves of the anomaly pixels and the background pixels in the input image and the reconstructed image, it is found that the difference of the spectral curves between the input and reconstructed anomaly pixels is more significant than the difference of the spectral curves between the input and reconstructed background pixels, as shown in Fig. 1. The SAD depicts the similarity of the two spectral curves. A small SAD value indicates that the two spectral curves are similar.

There are two main reasons for this phenomenon. First, considering that the distribution of the background is Gaussian, the Gaussian constraint is applied to the latent layer. Thus, the background is reconstructed well. Then, the total number of background pixels is far more than the number of anomaly pixels in the input data since the anomalies are small areas and the backgrounds occupy most of the area in a HSI in general. With sufficient background samples, the SLDR model would produce low reconstruction errors for background samples compared to anomaly samples. The similarity between the input and the reconstructed background pixels must be higher than that of the anomaly pixels.

A. Data Preparation and Introduction for SLDR Model

We begin with the introduction of the hyperspectral data preparation and other notation for the SLDR model. We denote the input set with $N_1 \times N_2$ spectral vectors as $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_i, \ldots, \mathbf{x}_{N_1 \times N_2}\}$, where $\mathbf{x}_i$, $(i = 1, 2, \ldots, N_1 \times N_2)$ is represented by the B-dimensional spectral vector of the ith pixel; N_1 and N_2 are the width and height of the HSI, respectively; and B is the number of spectral bands. The $N_1 \times N_2$ B-dimensional spectral vectors are input into the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:46:56 UTC from IEEE Xplore. Restrictions apply.

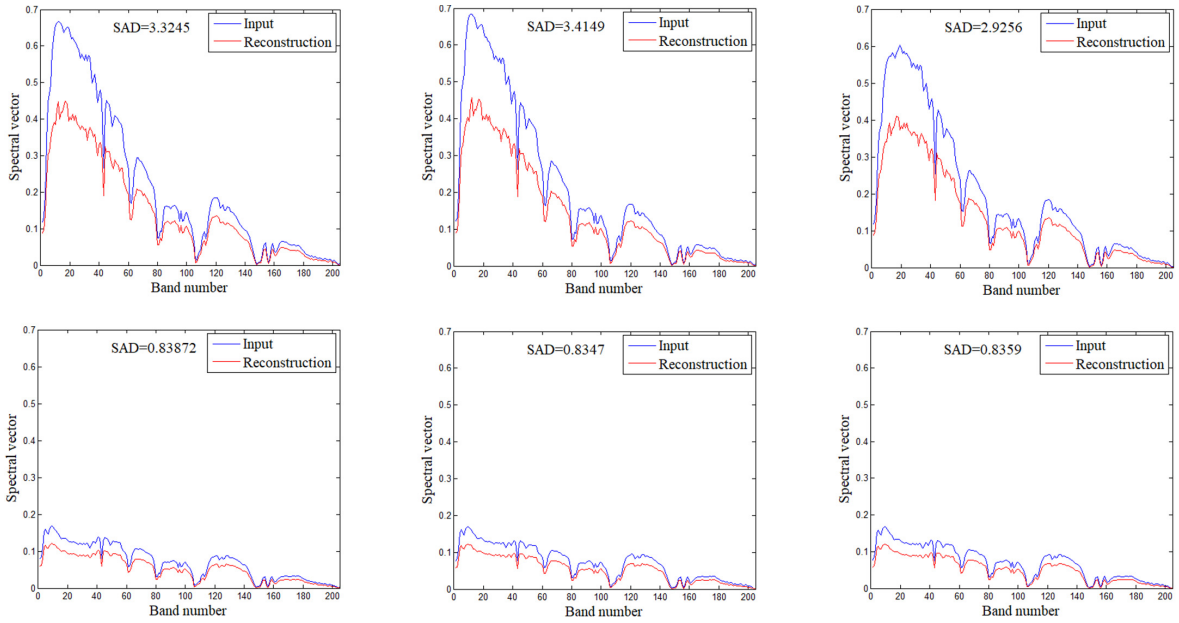


Fig. 1. Spectral curves of anomaly pixels and background pixels. First line: spectral curves of the anomalous pixels. Second line: spectral curves of the background pixels.

SLDR model. $\mathbf{Z} = \{\mathbf{z}_1, \ldots, \mathbf{z}_i, \ldots, \mathbf{z}_{N_1 \times N_2}\}$ is the encoded data in the latent layer, where $\mathbf{z}_i$ is the D-dimensional spectral vector of the ith pixel. $\hat{\mathbf{X}} = \{\hat{\mathbf{x}}_1, \ldots, \hat{\mathbf{x}}_i, \ldots, \hat{\mathbf{x}}_{N_1 \times N_2}\}$ is the data that are reconstructed by the SLDR model, where $\hat{\mathbf{x}}_i$ is the B-dimensional spectral vector of the ith pixel.

B. Model Architecture

Fig. 2 shows the diagram of the proposed method, which consists of the following steps. First, the input $\mathbf{x}_i \in \mathbb{R}^B$ is input into the encoder network, which maps the input into a latent space to obtain a latent vector $\mathbf{z}_i \in \mathbb{R}^D$ with the mapping function

$$\mathbf{z}_i = \sigma(\mathcal{W}_e \mathbf{x}_i + b_e) \tag{5}$$

where B is the dimension of an observation, and D is the dimension of a latent variable, $\mathcal{W}_e$ is the encoder weight, b_e is bias, $\sigma(\cdot)$ indicates the activation function. The activation function used in the encoder network is leaky ReLu (LReLu) [54]. We set two hidden layers in the network, where the number of nodes in the second hidden layer is set to 20 as $D = 20$. In this process, imposing a constraint on the encoder forces it to generate latent variables that obey a unit Gaussian distribution. Next, the latent vector $\mathbf{z}_i$ is decoded to reconstruct the input by using a decoder network with the mapping function

$$\hat{\mathbf{x}}_i = \sigma(\mathcal{W}_d \mathbf{z}_i + b_d) \tag{6}$$

where $\mathcal{W}_d$ is the decoder weight, b_d is bias, $\sigma(\cdot)$ indicates the activation function. The activation function used in the decoder network is also LReLu. Then, the SEM $\mathbf{S} = \{\mathrm{SAD}(\mathbf{x}_i, \hat{\mathbf{x}}_i)\} \in \mathbb{R}^{N_1 \times N_2}, (i = 1, \ldots, N_1 \times N_2)$ between the input and the reconstructed input is obtained by the spectral angle mapper (SAM), where $\mathrm{SAD}(\mathbf{x}_i, \hat{\mathbf{x}}_i)$ will be introduced in the next paragraph. Since the opening and closing operations in an attribute filter can remove the bright and dark connected components in small areas, respectively [55], the SEM $\mathbf{S}$ is input into an attribute filter to preserve the small area anomalies to get the initial anomaly detection results $\mathbf{Y}_1 \in \mathbb{R}^{N_1 \times N_2}$. After that, refined detection is carried out, where $\mathbf{X}' \in \mathbb{R}^{N_1 \times N_2 \times B}$ is obtained by

$$\mathbf{X}'_k = (1 - e^{-\beta \mathbf{Y}_1}) \cdot \mathbf{X}_k \tag{7}$$

where $\mathbf{X}'_k \in \mathbb{R}^{N_1 \times N_2}, (k = 1, \ldots, B)$ is the kth band of $\mathbf{X}'$, $\mathbf{X}_k \in \mathbb{R}^{N_1 \times N_2}$ is the kth band of $\mathbf{X}$. The specific instructions will be described in Section III-D. The background in $\mathbf{X}'$ is better suppressed and the anomaly is more prominent. Finally, the Mahalanobis distance is utilized to detect the anomalous pixels in $\mathbf{X}'$ to get the final anomaly detection results $\mathbf{Y}_2 \in \mathbb{R}^{N_1 \times N_2}$.

In the SLDR model, the SAM calculates the SAD of the spectral vector of each pixel in the HSI between the input and the reconstruction, as shown in the SEM $\mathbf{S}$ in Fig. 2. The formula for the SAD is defined as

$$\mathrm{SAD}(\mathbf{x}_i, \hat{\mathbf{x}}_i) = \cos^{-1}(\theta(\mathbf{x}_i, \hat{\mathbf{x}}_i)) \tag{8}$$

where

$$\theta(\mathbf{x}_i, \hat{\mathbf{x}}_i) = \frac{\mathbf{x}_i \hat{\mathbf{x}}_i}{\|\mathbf{x}_i\|_2 \|\hat{\mathbf{x}}_i\|_2}. \tag{9}$$

The SAD of the background pixels is far less than the SAD of the anomaly pixels between the input and the reconstruction. Hence, the background is suppressed and the anomaly is highlighted in the SEM. The performance of the anomaly detection can be improved by subjecting the SEM to an anomaly detector. To further improve the detection performance, it is necessary to reconstruct the backgrounds better. In this way, the SAD of the anomaly pixels between the input and the reconstruction will increase, and the SAD of the background pixels will decrease, which will make the anomalies more

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:46:56 UTC from IEEE Xplore. Restrictions apply.

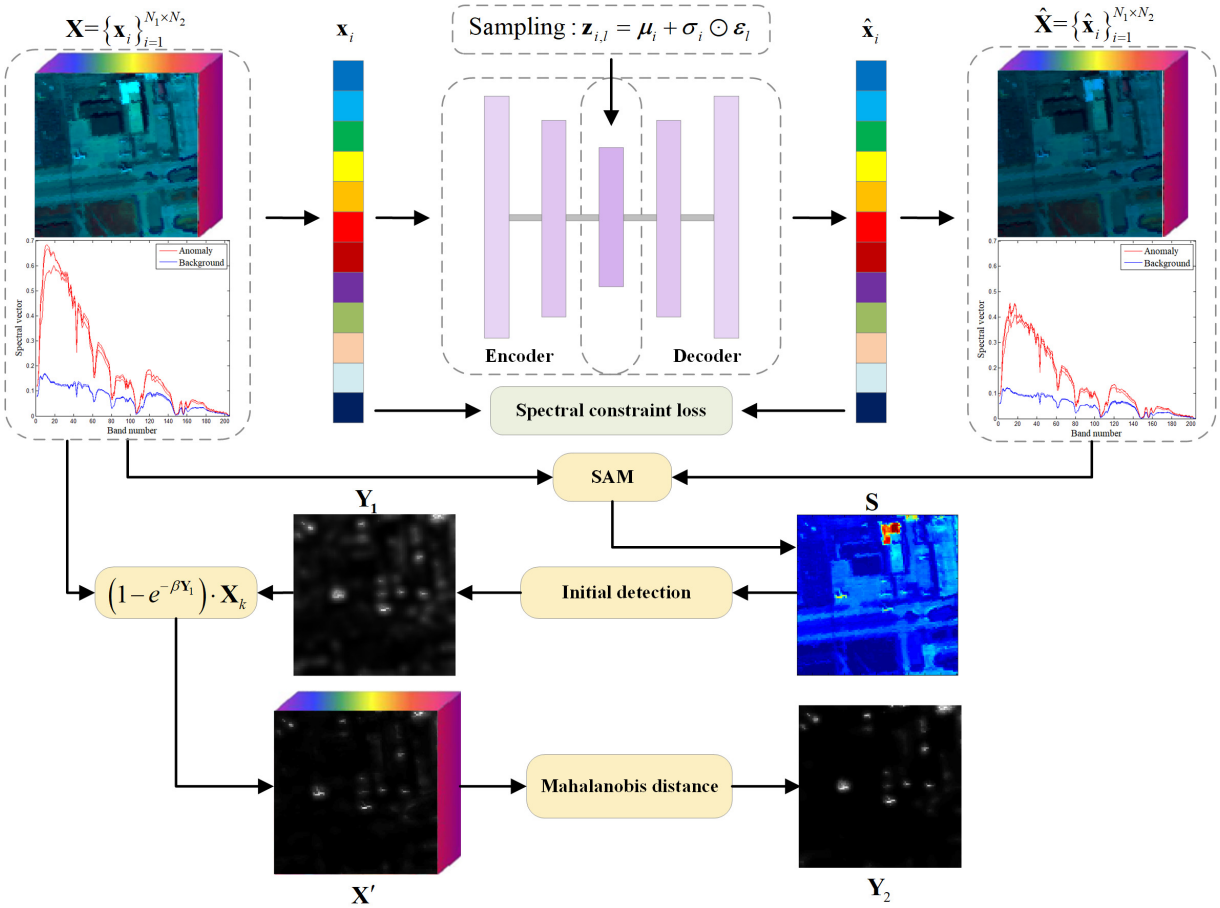


Fig. 2. Diagram of the proposed SLDR anomaly detection method.

obvious. The following is a detailed explanation of how to design a model that can have a good effect when reconstructing the background.

C. Loss Function of SLDR Model

Assuming that the samples in the input database have some kind of potential distribution, it is expected that the distribution can be learned by the proposed network. According to the prior knowledge that the background of HSIs obeys a Gaussian distribution, we expect the SLDR model to learn a Gaussian distribution well. In addition, a HSI contains hundreds of spectral bands, which provide important spectral information. The key to excellent performance when reconstructing the background is to make full use of the spectral information. Therefore, the loss function of the proposed SLDR model is defined as

$$\mathcal{L}_{\text{SLDR}} = \frac{\lambda_1}{n}\sum_{i=0}^{n}\|\mathbf{x}_i-\hat{\mathbf{x}}_i\|_2^2+\lambda_2 D_{\text{KL}}[q_\phi(\mathbf{z}|\mathbf{x})\|p_\theta(\mathbf{z})] + \lambda_3\sum_{i=0}^{n}\text{S A D}(\mathbf{x}_i,\hat{\mathbf{x}}_i) \quad (10)$$

where $n = N_1 \times N_2$ is the number of sample points.

The first term of the loss function is the reconstruction error of the input, which can be measured by the mean square error (MSE). MSE is the square of the Euclidean (ℓ_2) norm. This item uses the loss function in the AE [43]. λ_1 is the parameter that controls the influence of the MSE.

Because we constrain the encoder to force it to produce latent variables that obey a unit Gaussian distribution, the second term of the loss function uses the KL divergence to quantify the difference between the distribution of the latent variables and the unit Gaussian distribution, which refers to the loss function of the VAE [46]. The parameter λ_2 determines the effect of the Gaussian distribution's goodness of fit.

For a given HSI scene, the vast majority of pixels are background pixels. To reconstruct the background better, it is necessary to make the spectra between the original and the reconstructed data more similar. However, using the Euclidean (ℓ_2) norm and the KL divergence as the loss function to minimize the error between the input samples and their reconstructed versions cannot completely solve the problem. We introduce an additional penalty term, the SAD, as the third term in the loss function to obtain more discriminative latent abstracts and the angular similarity between the input and the reconstruction. The SAD measures a similarity score of two samples, $\mathbf{x}_i$ and $\hat{\mathbf{x}}_i$, and a score closer to zero implies a higher correlation. In addition, λ_3 is the parameter that controls the effect of the SAD.

D. Refined Detection

In the preliminary detection result $\mathbf{Y}_1$, the value of anomalous pixels is relatively larger, while the value of background

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:46:56 UTC from IEEE Xplore. Restrictions apply.

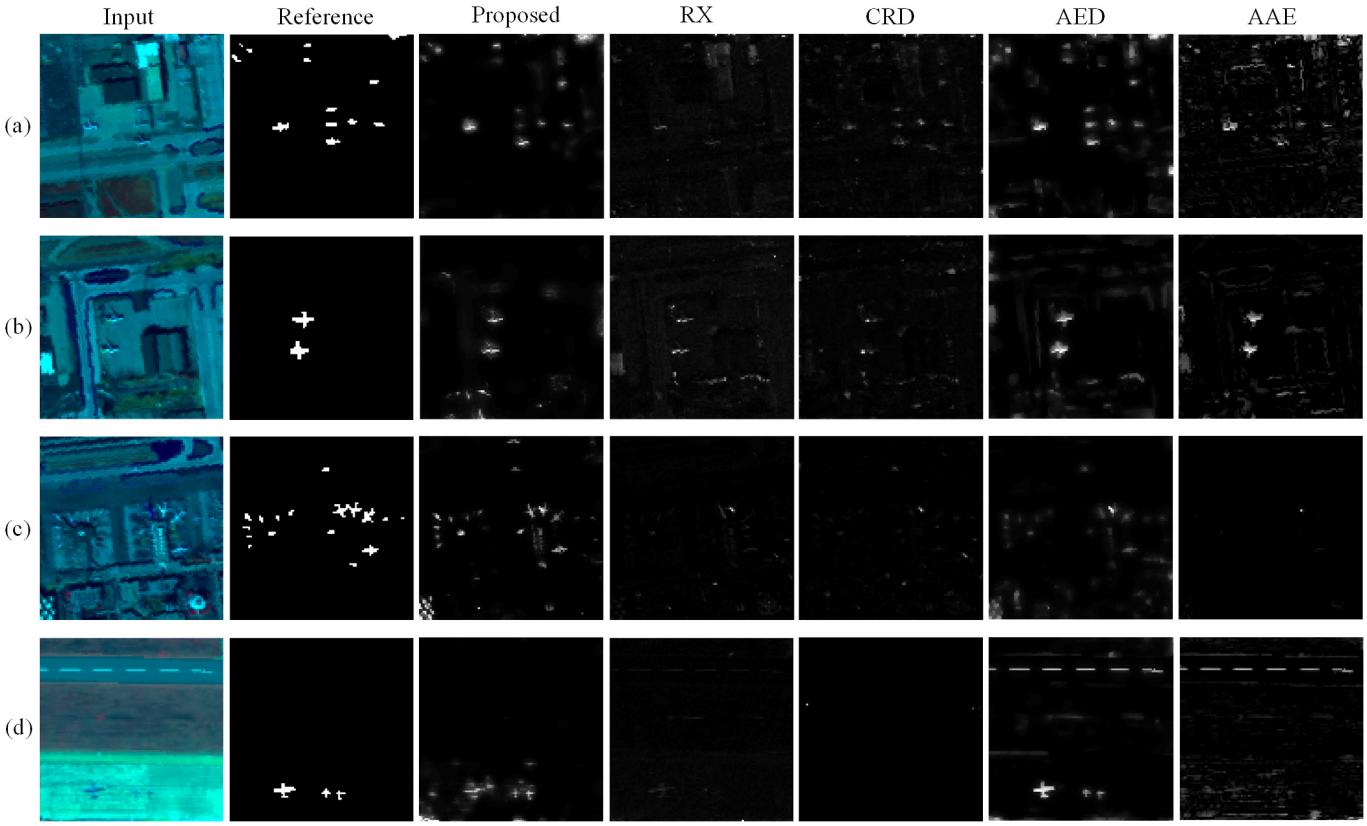


Fig. 3. Color composites of HSIs and detection maps for the ABU-airport scene. Second column: reference detection maps, which are obtained by human labeling. From left to right: false color image, reference, RX, CRD, AED, and AAE, respectively.

TABLE I
AUC SCORES OF (P_d, P_f) ON ABU-AIRPORT SCENE (FIG. 3)

Image	Proposed	RX	CRD	AED	AAE
a	0.98556	0.82259	0.96071	**0.99226**	0.87469
b	0.98541	0.84059	0.92071	**0.99360**	0.96977
c	**0.98497**	0.92899	0.94722	0.97562	0.92195
d	0.99468	0.95206	0.82168	**0.99534**	0.94410
Average	0.98766	0.88606	0.91258	**0.98920**	0.92763

TABLE II
AUC SCORES OF (P_d, P_f) ON ABU-BEACH SCENE (FIG. 4)

Image	Proposed	RX	CRD	AED	AAE
a	**0.99860**	0.98281	0.96720	0.99740	0.95365
b	**0.99411**	0.91060	0.92214	0.95495	0.99083
c	0.99946	0.99985	0.99914	**0.99968**	0.99941
d	**0.99940**	0.95335	0.90093	0.97935	0.97961
Average	**0.99789**	0.96165	0.94735	0.98285	0.98088

TABLE III
AUC SCORES OF (P_d, P_f) ON ABU-URBAN SCENE (FIG. 5)

Image	Proposed	RX	CRD	AED	AAE
a	**0.99878**	0.99064	0.99279	0.99815	0.98922
b	**0.99921**	0.99463	0.93042	0.78570	0.96806
c	0.98047	0.95119	0.93970	**0.99754**	0.99211
d	0.99166	0.98870	0.95156	0.99116	**0.99489**
e	**0.99437**	0.96932	0.93045	0.98452	0.98026
Average	**0.99290**	0.97890	0.94898	0.95141	0.98491

pixels is close to 0. Therefore, we construct a weight function with y_1 as an independent variable

$$w(y_1) = (1 - e^{-\beta y_1}) \tag{11}$$

where y_1 is the pixel in the preliminary detection result $\mathbf{Y}_1$. The range of $w(y_1)$ is from 0 to 1. When y_1 has a large value, $w(y_1)$ will tend to stably approach 1, whereas $w(y_1)$ will sharply drop almost close to 0. Finally, a weight matrix $\mathcal{W}$ is obtained. For further refined detection, we take the dot product between the weight matrix $\mathcal{W}$ and each band of the input HSI $\mathbf{X}_k$ as

$$\mathbf{X}'_k = \mathcal{W} \cdot \mathbf{X}_k = (1 - e^{-\beta \mathbf{Y}_1}) \cdot \mathbf{X}_k. \tag{12}$$

If pixel y_1 has a large value in $\mathbf{Y}_1$, its corresponding pixel in $\mathbf{X}'_k$ will get almost the same value as in $\mathbf{X}_k$. If pixel y_1 has a small value in $\mathbf{Y}_1$, its corresponding pixel in $\mathbf{X}_k$ will be multiplied by a value approaching 0 to be suppressed. As shown in Fig. 2, the background in $\mathbf{X}'$ is suppressed very

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:46:56 UTC from IEEE Xplore. Restrictions apply.

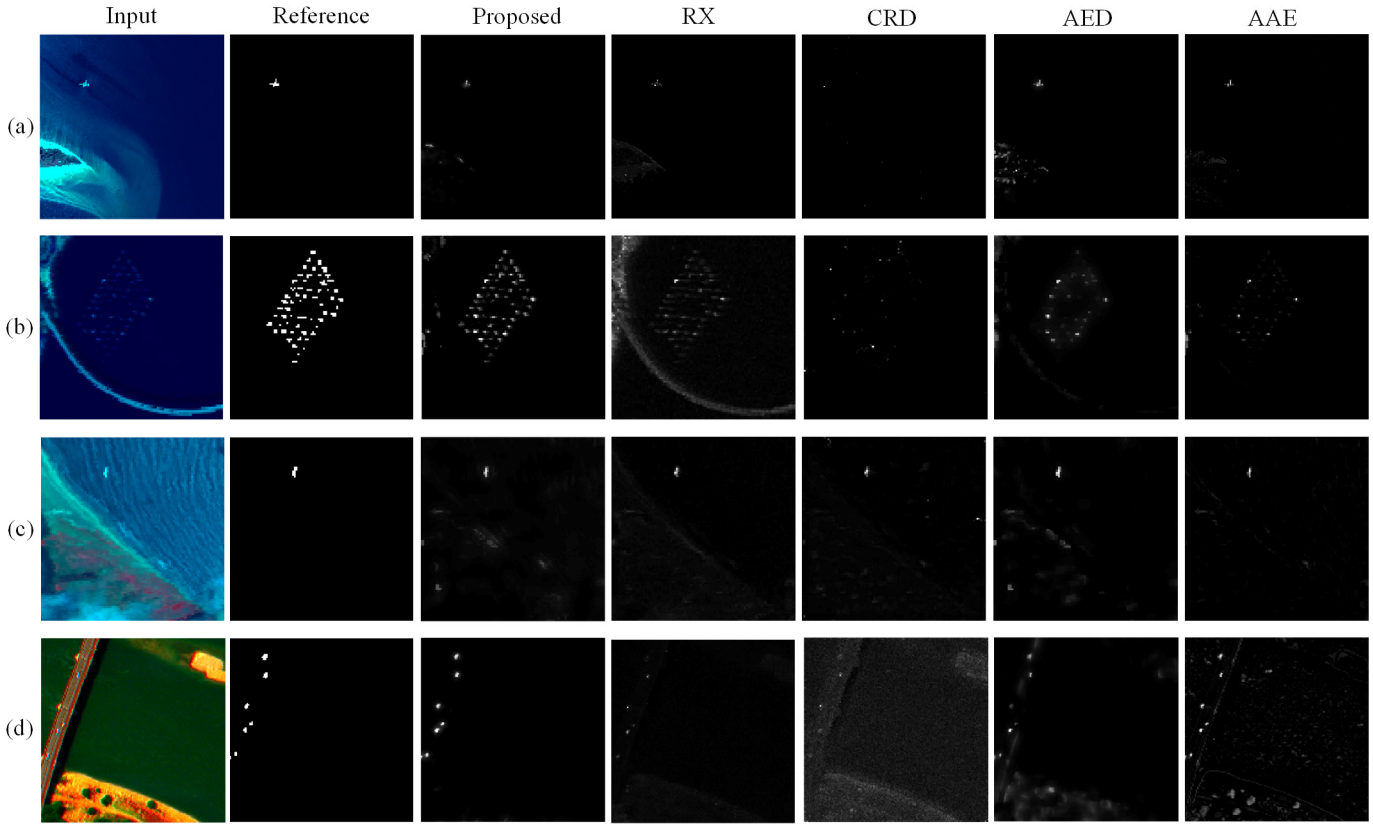


Fig. 4. Color composites of HSIs and detection maps for the ABU-beach scene. Second column: reference detection maps, which are obtained by human labeling. From left to right: false color image, reference, RX, CRD, AED, and AAE, respectively.

TABLE IV
AUC SCORES OF (P_d, P_f) ON SAN DIEGO AIRPORT (FIG. 6)

Image	Proposed	RX	CRD	AED	AAE
San Diego	**0.99253**	0.95151	0.83742	0.97966	0.95946

TABLE V
AUC SCORES OF (P_f, τ) ON ABU-AIRPORT SCENE (FIG. 3)

Image	Proposed	RX	CRD	AED	AAE
a	**0.02028**	0.04232	0.03362	0.03293	0.03451
b	0.02325	0.05154	0.02964	0.03011	**0.01609**
c	0.01557	0.01444	**0.01187**	0.02249	0.01361
d	0.01728	0.02471	**0.00079**	0.02425	0.02765
Average	0.01910	0.03325	**0.01898**	0.02744	0.02297

well, and the distinction between the anomaly and the background is particularly evident. Therefore, we further utilize the Mahalanobis distance for refined anomalous detection based on the great separation characteristics of the background and the anomaly in $\mathbf{X}'$. The detection result can be represented as follows:

$$y_2 = \left(\mathbf{x}_i' - \boldsymbol{\mu}\right)^T \boldsymbol{\Gamma}^{-1} \left(\mathbf{x}_i' - \boldsymbol{\mu}\right) \tag{13}$$

TABLE VI
AUC SCORES OF (P_f, τ) ON ABU-BEACH SCENE (FIG. 4)

Image	Proposed	RX	CRD	AED	AAE
a	0.00231	0.00652	**0.00076**	0.00410	0.00296
b	0.00850	0.04875	0.00282	0.01227	**0.00121**
c	0.01818	0.02592	0.01935	0.01161	**0.00589**
d	**0.00318**	0.02332	0.11240	0.01998	0.01941
Average	0.00804	0.02613	0.03383	0.01199	**0.00737**

where y_2 is the pixel in the refined detection result $\mathbf{Y}_2$, $\mathbf{x}_i'$, $(i = 1, 2, \ldots, N_1 \times N_2)$ is represented by the B-dimensional spectral vector of the ith pixel, and $\boldsymbol{\mu}$ and $\boldsymbol{\Gamma}$ are the mean vector and the covariance matrix of $\mathbf{X}'$, respectively. After calculating the Mahalanobis distance of each spectral vector, the value of y_2 obtained from anomaly pixels will be larger than that of the background pixels. The reason is that by suppressing the background, the distance between the spectral vectors of abnormal pixels and the mean vector is increased.

IV. EXPERIMENT

A. Hyperspectral Data Sets

In this section, two hyperspectral data sets that were captured at different scenes are used to evaluate the performance of the proposed algorithm.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:46:56 UTC from IEEE Xplore. Restrictions apply.

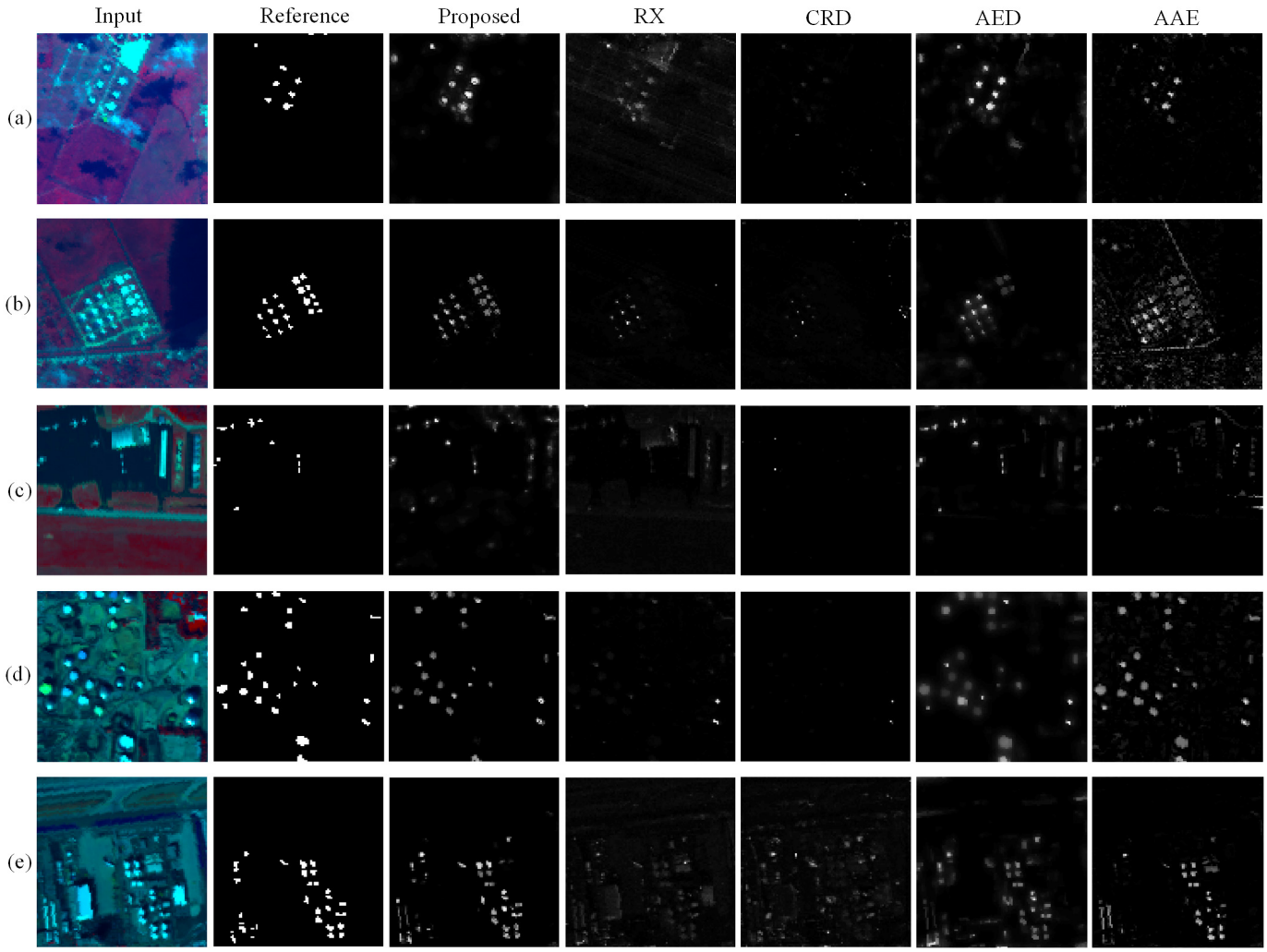


Fig. 5. Color composites of HSIs and detection maps for the ABU-urban scene. Second column: reference detection maps, which are obtained by human labeling. From left to right: false color image, reference, RX, CRD, AED, and AAE, respectively.

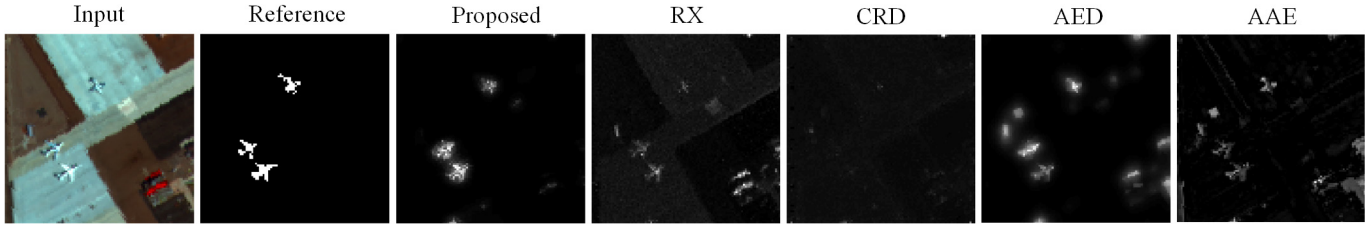


Fig. 6. Color composites of HSIs and detection maps for the San Diego scene. Second column: reference detection maps, which are obtained by human labeling.

1) Airport-Beach-Urban (ABU) Data Set: The first and second columns of Figs. 3–5 are the sample images and reference detection maps of the ABU data set, respectively. The ABU data set was made available online.[1] These sample images contain airport, beach, and urban scenes, and most of them were captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. Only one of the beach scenes is captured by the Reflective Optics System Imaging Spectrometer (ROSIS-03) Sensor. The size of each band is 100×100 or 150×150 pixels. The detailed description is in [56].

2) San Diego Data Set: The sample image and reference detection map of the second data set are, respectively, shown in the first and second columns of Fig. 6. It was captured by the AVIRIS sensor over the San Diego airport area, CA, USA. The image size is $100 \times 100 \times 189$, and the noisy bands have been removed. The two planes are regarded as anomaly objects in the image.

[1] http://xudongkang.weebly.com/

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:46:56 UTC from IEEE Xplore. Restrictions apply.

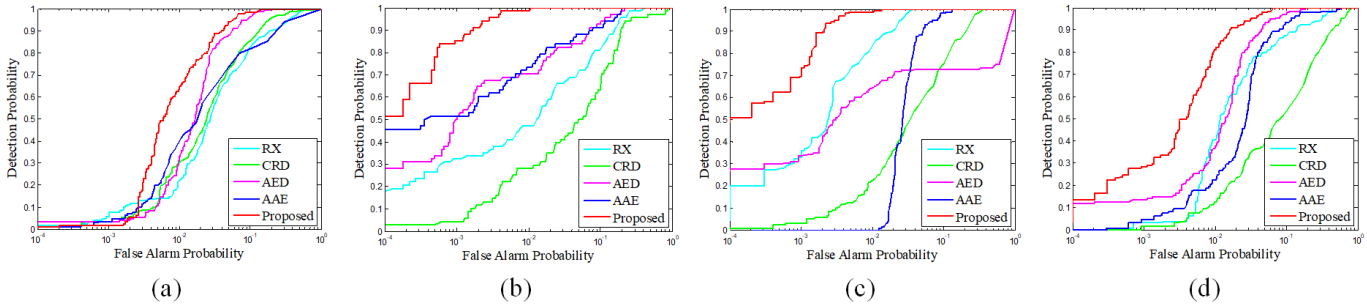


Fig. 7. ROC curves of (P_d, P_f) obtained by the compared methods on (a) third airport scene in ABU-Airport data set, (b) fourth beach scene in ABU-Beach data set, (c) second urban scene in ABU-Urban data set, and (d) San Diego.

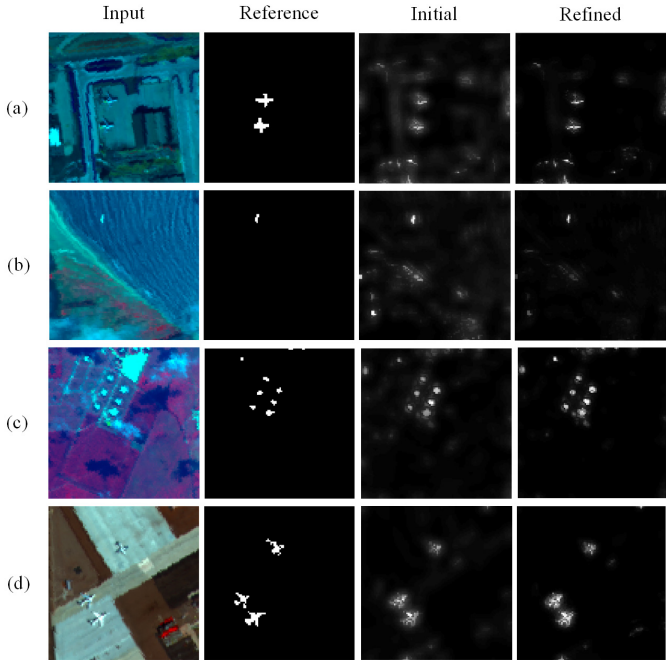


Fig. 8. Detection result maps obtained by the main components. From left to right: false color image, reference, initial result, and refined result, respectively.

TABLE VII
AUC SCORES OF (P_f, τ) ON ABU-URBAN SCENE (FIG. 5)

Image	Proposed	RX	CRD	AED	AAE
a	0.01807	0.05552	**0.00889**	0.02292	0.00941
b	**0.00444**	0.01351	0.01449	0.01437	0.04229
c	0.02037	0.03502	**0.00229**	0.01379	0.00742
d	0.00392	0.01140	**0.00121**	0.03069	0.02030
e	**0.00677**	0.04370	0.04447	0.03058	0.00915
Average	**0.01071**	0.03183	0.01427	0.02247	0.01771

TABLE VIII
AUC SCORES OF (P_f, τ) ON SAN DIEGO AIRPORT (FIG. 6)

Image	Proposed	RX	CRD	AED	AAE
San Diego	**0.01407**	0.07037	0.05874	0.03441	0.02841

TABLE IX
EVALUATION AUC SCORES OF THE MAIN COMPONENTS OF PROPOSED METHODS ON ABU-AIRPORT SCENE

Image	(P_d, P_f)		(P_f, τ)	
	Initial result	Refined result	Initial result	Refined result
a	**0.98809**	0.98556	0.06456	**0.02028**
b	0.98370	**0.98541**	0.07268	**0.02325**
c	0.98388	**0.98497**	0.04068	**0.01557**
d	0.99102	**0.99468**	0.03896	**0.01728**
Average	0.98667	**0.98766**	0.05422	**0.01910**

B. Detection Performance

To evaluate the anomaly detection performance of the proposed SLDR, we use three widely anomaly detection methods, that is, the RX detector [23], CRD [34], AED [56], and AAE [57] as the comparison algorithms. These methods are often cited in the related literature.

1) Evaluation Criteria: We use two evaluation indexes (the receiver operating characteristic (ROC) curves [58] and the area under the ROC curves (AUC) [59]) to analyze and compare the anomaly detection performance of different algorithms. Detection probability (P_d), false alarm probability (P_f), and threshold (τ) can be applied to generate three types of ROC curves, that is, the ROC curve of (P_d, P_f), the ROC curve of (P_d, τ), and the ROC curve of (P_f, τ). The ROC curve of (P_d, P_f) indicates the tradeoff between the P_d and P_f. The ROC curve of (P_f, τ) is designed to assess the degree of background suppression. The AUC score is the area that is surrounded by the coordinate axis under the ROC curve. It is used as an evaluation criterion because, in many cases, the ROC curve cannot accurately represent which detector is more effective. Detectors with higher AUC scores of (P_d, P_f) have better detection performance. And detectors with lower AUC scores of (P_f, τ) have better background suppression.

2) Detection Performance: In this section, the detection performance of the proposed SLDR algorithm is evaluated and compared with four conventional anomaly detection algorithms, which are the RX, CRD, AED, and AAE. Experiments are carried out on the ABU and San Diego data sets. The RX detector is classical Mahalanobis distance-based anomaly detection algorithms. The CRD detector also has leading

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:46:56 UTC from IEEE Xplore. Restrictions apply.

TABLE X
EVALUATION AUC SCORES OF THE MAIN COMPONENTS OF PROPOSED METHODS ON ABU-BEACH SCENE

Image	(P_d, P_f)		(P_f, τ)	
	Initial result	Refined result	Initial result	Refined result
a	0.99292	**0.99860**	0.02471	**0.00231**
b	**0.99568**	0.99411	**0.00581**	0.00850
c	0.99901	**0.99946**	0.06668	**0.01818**
d	0.97941	**0.99940**	0.02713	**0.00318**
Average	0.99175	**0.99789**	0.03108	**0.00804**

TABLE XI
EVALUATION AUC SCORES OF THE MAIN COMPONENTS OF PROPOSED METHODS ON ABU-URBAN SCENE

Image	(P_d, P_f)		(P_f, τ)	
	Initial result	Refined result	Initial result	Refined result
a	0.99837	**0.99878**	0.04477	**0.01807**
b	**0.99947**	0.99921	0.01312	**0.00444**
c	0.98044	**0.98047**	0.03400	**0.02037**
d	**0.99368**	0.99166	0.00599	**0.00392**
e	0.99405	**0.99437**	0.02549	**0.00677**
Average	**0.99320**	0.99290	0.02467	**0.01071**

performance in hyperspectral anomaly detection, and the AED is a recently proposed algorithm with high detection accuracy. The AAE is proposed in [57]. To ensure the rationality of the experiment, each algorithm is adjusted to the optimal parameters. Because the detection performance of the CRD is insensitive to the value of the regularization parameter λ, according to the recommendation of Li and Du [34], the value of this parameter is set to 10^{-6}. For the AED method, we obtain the detection result of each data set according to the parameters and code given in [56].

First, the 2-D grayscale maps show the sample images, the reference detection maps, and the detection maps of the comparison methods and the proposed anomaly detection algorithm in Figs. 3–6. The visual detection results of the RX and the CRD are that some anomaly pixels can be detected, but some anomaly pixels are missed. The AED and the AAE can detect most of the anomaly pixels, but they also incorrectly detect the background pixels as anomaly pixels. Compared with RX, CRD, AED, and AAE, the detection results of the proposed method are close to the reference map for each data set. It is evident that the results of the SLDR demonstrate better performance and higher confidence than the other detection algorithms.

Moreover, because visual inspection cannot obtain accurate estimates, we propose more quantitative evaluation criteria. Fig. 7 shows the ROC curves of (P_d, P_f) of the proposed SLDR algorithm and its competitors. As shown in Fig. 7, the RX and CRD perform terribly and generate high AUC scores of (P_f, τ) on two data sets because of failing to suppress the backgrounds. The ROC curves of (P_d, P_f) of the SLDR algorithm are almost always above those of the other algorithms. Compared with the other methods, the SLDR can reasonably control low AUC scores of (P_f, τ) while also detecting anomalies. The AUC scores of (P_d, P_f) of each method are presented in Tables I–IV, and the AUC scores of (P_f, τ) of each method are presented in Tables V–VIII. The AUC scores of (P_d, P_f) of the SLDR are higher than those of other algorithms on most tested HSIs. For the ABU data sets, the average AUC scores of (P_d, P_f) that are obtained by the SLDR method are 0.98766, 0.99789, and 0.99290, respectively, which are significantly higher than those obtained by the RX, CRD, and AAE. The average AUC scores of (P_d, P_f) of both the ABU-Beach scene and the ABU-Urban scene are higher than those of the AED, and the average AUC scores of (P_f, τ) are 0.00804 and 0.01071, respectively, which are lower than the AED. Although the average AUC score of (P_d, P_f) of the ABU-Airport scene is slightly below that of the AED, the corresponding AUC score of (P_f, τ) is 0.01910, which is lower than that of the AED. For the San Diego data set, the AUC score of (P_d, P_f) is 0.99253, which is highest compared to other algorithms. As seen from Fig. 7 and Tables I–VIII, the proposed SLDR algorithm is more effective than other detection methods.

TABLE XII
EVALUATION AUC SCORES OF THE MAIN COMPONENTS OF PROPOSED METHODS ON SAN DIEGO AIRPORT

Image	(P_d, P_f)		(P_f, τ)	
	Initial result	Refined result	Initial result	Refined result
San Diego	0.98410	**0.99253**	0.03516	**0.01407**

3) Component Analysis: To further validate the effect of refined detection, we provide the objective and subjective results of initial detection and refined detection. The detection maps of four images are selected and displayed in Fig. 8. The four images contain different structures and different proportion of anomalies, which are also captured over different scenes. As can be seen from the third column of Fig. 8, the initial detection can detect almost all anomalies, but some background pixels are mistakenly regarded as anomalies. After refined detection, the background is better suppressed, as shown in the fourth column of Fig. 8. The AUC scores based on the above two steps are presented in Tables IX–XII. It is apparent that the AUC scores of (P_f, τ) obtained by refined detection are greatly reduced, and the AUC scores of (P_d, P_f) are slightly improved. Consequently, each component of our method has a positive impact on the performance of anomaly detection.

V. CONCLUSION

In this article, a novel discriminative reconstruction with spectral learning (SLDR) method is proposed for the anomaly detection of HSIs. To better suppress the background and highlight the anomalies, we first impose a constraint on the encoder to force it to generate latent variables that obey a unit Gaussian distribution, considering that the distribution of the background is Gaussian. Then, the loss function of the proposed SLDR model additionally introduces the SAD term,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:46:56 UTC from IEEE Xplore. Restrictions apply.

which constrains the model to generate a reconstruction having greater spectral similarity to the input. Finally, we use the SEM to detect anomalies, in which the anomalies are obviously separate from the background. Experiments on the two real HSI data sets demonstrate that our proposed algorithm can achieve excellent detection performance. Compared with RX, CRD, AED, and AAE, the results show that the proposed SLDR method outperforms other classical anomaly detection algorithms.

REFERENCES

[1] L. Zhu and G. Wen, "Hyperspectral anomaly detection via background estimation and adaptive weighted sparse representation," *Remote Sens.*, vol. 10, no. 2, p. 272, 2018.

[2] Y. Yuan, X. Zheng, and X. Lu, "Spectral–spatial kernel regularized for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3815–3832, Jul. 2015.

[3] L. Zhao, W. Lin, Y. Wang, and X. Li, "Recursive local summation of RX detection for hyperspectral image using sliding windows," *Remote Sens.*, vol. 10, no. 2, p. 103, 2018.

[4] Q. Wang, J. Lin, and Y. Yuan, "Salient band selection for hyperspectral image classification via manifold ranking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 6, pp. 1279–1289, Jun. 2016.

[5] A. Alarcon-Ramirez, M. R. Rwebangira, M. F. Chouikha, and V. Manian, "A new methodology based on level sets for target detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 9, pp. 5385–5396, Sep. 2016.

[6] Y. Zhang, K. Wu, B. Du, L. Zhang, and X. Hu, "Hyperspectral target detection via adaptive joint sparse representation and multi-task learning with locality information," *Remote Sens.*, vol. 9, no. 5, p. 482, 2017.

[7] D. W. J. Stein, S. G. Beaven, L. E. Hoff, E. M. Winter, A. P. Schaum, and A. D. Stocker, "Anomaly detection from hyperspectral imagery," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 58–69, Jan. 2002.

[8] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2019.

[9] L. Fang, G. Liu, S. Li, P. Ghamisi, and J. A. Benediktsson, "Hyperspectral image classification with squeeze multibias network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1291–1301, Mar. 2019.

[10] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[11] K.-H. Liu, S.-Y. Chen, H.-C. Chien, and M.-H. Lu, "Progressive sample processing of band selection for hyperspectral image transmission," *Remote Sens.*, vol. 10, no. 3, p. 367, 2018.

[12] Q. Wang, Q. Li, and X. Li, "Hyperspectral band selection via adaptive subspace partition strategy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4940–4950, Dec. 2019.

[13] X. Zhang *et al.*, "Hyperspectral unmixing via low-rank representation with space consistency constraint and spectral library pruning," *Remote Sens.*, vol. 10, no. 2, p. 339, 2018.

[14] M. Rizkinia and M. Okuda, "Joint local abundance sparse unmixing for hyperspectral images," *Remote Sens.*, vol. 9, no. 12, p. 1224, 2017.

[15] D. Manolakis and G. Shaw, "Detection algorithms for hyperspectral imaging applications," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 29–43, Jan. 2002.

[16] T. Jiang, Y. Li, W. Xie, and Q. Du, "Discriminative reconstruction constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, early access, Jan. 28, 2019, doi: 10.1109/TGRS.2020.2965961.

[17] G. Shaw and D. Manolakis, "Signal processing for hyperspectral image exploitation," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 12–16, Jan. 2002.

[18] L. Zhang, L. Zhang, D. Tao, X. Huang, and B. Du, "Hyperspectral remote sensing image subpixel target detection based on supervised metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4955–4965, Aug. 2014.

[19] L. Zhang, L. Zhang, D. Tao, and X. Huang, "Sparse transfer manifold embedding for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 2, pp. 1030–1043, Feb. 2014.

[20] N. M. Nasrabadi, "Hyperspectral target detection: An overview of current and future challenges," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 34–44, Jan. 2014.

[21] Y. Yuan, D. Ma, and Q. Wang, "Hyperspectral anomaly detection by graph pixel selection," *IEEE Trans. Cybern.*, vol. 46, no. 12, pp. 3123–3134, Dec. 2016.

[22] Y. Qu *et al.*, "Hyperspectral anomaly detection through spectral unmixing and dictionary-based low-rank decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4391–4405, Aug. 2018.

[23] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[24] M. Imani, "RX anomaly detector with rectified background," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1313–1317, Aug. 2017.

[25] M. Huber-Lerner, O. Hadar, S. R. Rotman, and R. Huber-Shalem, "Hyperspectral band selection for anomaly detection: The role of data Gaussianity," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 2, pp. 732–743, Feb. 2016.

[26] W. Li, G. Wu, and Q. Du, "Transferred deep learning for anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 597–601, May 2017.

[27] J. M. Molero, E. M. Garzon, I. Garcia, and A. Plaza, "Analysis and optimizations of global and local versions of the RX algorithm for anomaly detection in hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 801–814, Apr. 2013.

[28] B. Du and L. Zhang, "Random-selection-based anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 5, pp. 1578–1589, May 2011.

[29] C.-I. Chang, "Target signature-constrained mixed pixel classification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 5, pp. 1065–1081, May 2002.

[30] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[31] S. Khazai and B. Mojaradi, "A modified kernel-RX algorithm for anomaly detection in hyperspectral images," *Arabian J. Geosci.*, vol. 8, no. 3, pp. 1487–1495, Mar. 2015.

[32] A. Schaum, "Joint subspace detection of hyperspectral targets," in *Proc. IEEE Aerosp. Conf.*, vol. 3, Mar. 2004, p. 1824.

[33] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Hyperspectral image classification via kernel sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 1, pp. 217–231, Jan. 2013.

[34] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[35] J. Li, H. Zhang, L. Zhang, and L. Ma, "Hyperspectral anomaly detection by the use of background joint sparse representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2523–2533, Jun. 2015.

[36] M. Dao, C. Kwan, B. Ayhan, and T. D. Tran, "Burn scar detection using cloudy MODIS images via low-rank and sparsity-based models," in *Proc. IEEE Global Conf. Signal Inf. Process. (GlobalSIP)*, Dec. 2016, pp. 177–181.

[37] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[38] W. Xie, T. Jiang, Y. Li, X. Jia, and J. Lei, "Structure tensor and guided filtering-based algorithm for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4218–4230, Jul. 2019.

[39] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Joint reconstruction and anomaly detection from compressive hyperspectral images using Mahalanobis distance-regularized tensor RPCA," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2919–2930, May 2018.

[40] Z. Wu, W. Zhu, J. Chanussot, Y. Xu, and S. Osher, "Hyperspectral anomaly detection via global and local joint modeling of background," *IEEE Trans. Signal Process.*, vol. 67, no. 14, pp. 3858–3869, Jul. 2019.

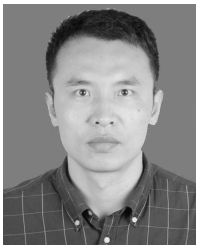
[41] Q. Wang, J. Wan, F. Nie, B. Liu, C. Yan, and X. Li, "Hierarchical feature selection for random projection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 5, pp. 1581–1586, May 2019.

[42] N. He, L. Fang, S. Li, A. Plaza, and J. Plaza, "Remote sensing scene classification using multilayer stacked covariance pooling," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 6899–6910, Dec. 2018.

[43] Y. Xia, X. Cao, F. Wen, G. Hua, and J. Sun, "Learning discriminative reconstructions for unsupervised outlier removal," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1511–1519.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:46:56 UTC from IEEE Xplore. Restrictions apply.

[44] L.-L. Cao, W.-B. Huang, and F.-C. Sun, "Building feature space of extreme learning machine with sparse denoising stacked-autoencoder," *Neurocomputing*, vol. 174, pp. 60–71, Jan. 2016.
[45] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," 2019, *arXiv:1901.03407*. [Online]. Available: http://arxiv.org/abs/1901.03407
[46] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*. [Online]. Available: http://arxiv.org/abs/1312.6114
[47] I. Goodfellow *et al.*, "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 2672–2680.
[48] J. An and S. Cho, "Variational autoencoder based anomaly detection using reconstruction probability," *Special Lect. IE*, vol. 2, no. 1, pp. 1–18, Dec. 2015.
[49] H. Zenati, C. Sheng Foo, B. Lecouat, G. Manek, and V. R. Chandrasekhar, "Efficient GAN-based anomaly detection," 2018, *arXiv:1802.06222*. [Online]. Available: http://arxiv.org/abs/1802.06222
[50] M. Kliger and S. Fleishman, "Novelty detection with GAN," 2018, *arXiv:1802.10560*. [Online]. Available: http://arxiv.org/abs/1802.10560
[51] J. Geweke, "Bayesian inference in econometric models using Monte Carlo integration," *Econometrica*, vol. 57, no. 6, pp. 1317–1339, Nov. 1989.
[52] Y. Lu and P. Xu, "Anomaly detection for skin disease images using variational autoencoder," 2018, *arXiv:1807.01349*. [Online]. Available: http://arxiv.org/abs/1807.01349
[53] B. Kiran, D. Thomas, and R. Parakkal, "An overview of deep learning based methods for unsupervised and semi-supervised anomaly detection in videos," *J. Imag.*, vol. 4, no. 2, p. 36, Feb. 2018.
[54] B. Xu, N. Wang, T. Chen, and M. Li, "Empirical evaluation of rectified activations in convolutional network," 2015, *arXiv:1505.00853*. [Online]. Available: http://arxiv.org/abs/1505.00853
[55] E. J. Breen and R. Jones, "Attribute openings, thinnings, and granulometries," *Comput. Vis. Image Understand.*, vol. 64, no. 3, pp. 377–389, Nov. 1996.
[56] X. Kang, X. Zhang, S. Li, K. Li, J. Li, and J. A. Benediktsson, "Hyperspectral anomaly detection with attribute and edge-preserving filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5600–5611, Oct. 2017.
[57] W. Xie, B. Liu, Y. Li, J. Lei, C.-I. Chang, and G. He, "Spectral adversarial feature learning for anomaly detection in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2352–2365, Apr. 2020.
[58] J. Kerekes, "Receiver operating characteristic curve confidence intervals and regions," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 2, pp. 251–255, Apr. 2008.
[59] C.-I. Chang and S.-S. Chiang, "Anomaly detection and classification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 6, pp. 1314–1325, Jun. 2002.



Jie Lei (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 2006 and 2010, respectively.

He has been a Visiting Scholar with the Department of Computer Science, University of California at Los Angeles, Los Angeles, CA, USA, from 2014 to 2015. He is an Associate Professor with the School of Telecommunications Engineering, Xidian University, and is a member of the Image Coding and Processing Center, State Key Laboratory of Integrated Services Networks. His research interests focus on image and video processing, computer vision, and customized computing for big-data applications.



Shuo Fang received the B.E. degree in communication engineering from Jilin University, Jilin, China, in 2018. She is pursuing the M.S. degree with the Image Coding and Processing Center, State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an, China.

Her research interests include hyperspectral image, anomaly detection, and unsupervised learning.



Weiying Xie (Member, IEEE) received the B.S. degree in electronic information science and technology from the University of Jinan, Jinan, China, in 2011, the M.S. degree in communication and information systems from Lanzhou University, Lanzhou, China, in 2014, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2017.

She is an Associate Professor with the State Key Laboratory of Integrated Services Networks, Xidian University. She has published more than 20 articles in refereed journals, including the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, *Neural Networks*, and *Pattern Recognition*. Her research interests include neural networks, machine learning, hyperspectral image processing, and high-performance computing.



Yunsong Li received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

He joined the School of Telecommunications Engineering, Xidian University, in 1999, where he is a Professor. He is the Director of the Image Coding and Processing Center, State Key Laboratory of Integrated Services Networks. His research interests focus on image and video processing and high-performance computing.



Chein-I Chang (Life Fellow, IEEE) received the B.S. degree in mathematics from Soochow University, Taipei, Taiwan, in 1973, the M.S. degree in mathematics from the Institute of Mathematics, National Tsing Hua University, Hsinchu, Taiwan, in 1975, the M.A. degree in mathematics from Stony Brook University, Stony Brook, NY, USA, in 1977, the M.S. and M.S.E.E. degrees from the University of Illinois at Urbana Champaign, Urbana, IL, USA, in 1980 and 1982, respectively, and the Ph.D. degree in electrical engineering from the University of Maryland at College Park, College Park, MD, USA, in 1987.

He has been a Distinguished Chair Professor with National Chung Hsing University, Taichung, Taiwan, since 2014, and a Chair Professor with Providence University, Taichung, since 2012. He has been with the University of Maryland, Baltimore County, Baltimore, MD, USA, since 1987, where he is currently a Professor with the Department of Computer Science and Electrical Engineering. He holds the Chang Jiang Scholar Chair Professorship and the Director of the Center for Hyperspectral Imaging in Remote Sensing, Dalian Maritime University, Dalian, China. He also holds the Hua Shan Scholar Chair Professorship with Xidian University, Xi'an, China. He holds seven patents in hyperspectral image processing. He has authored four books. He has edited two books and co-edited a book with A. Plaza. His research interests include multispectral/hyperspectral image processing, automatic target recognition, and medical imaging.

Dr. Chang is a fellow of SPIE. He was an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING from 2001 to 2007. He is currently an Associate Editor of *Artificial Intelligence Research* and also on the editorial boards of the JOURNAL OF HIGH SPEED NETWORKS, *Recent Patents on Mechanical Engineering*, the *International Journal of Computational Sciences and Engineering*, the *Journal of Robotics and Artificial Intelligence*, and the *Open Remote Sensing Journal*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:46:56 UTC from IEEE Xplore. Restrictions apply.